

Common Lung Disorders

Learning Objectives

- ⇒ To have an overall review of respiratory disorders
- ⇒ To know clinical features of pneumonia, chronic obstructive pulmonary diseases (COPD), pleural effusion, bronchiectasis
- ⇒ To know the physical examination routinely done in the examination of above said diseases
- ⇒ To know the different investigations performed and their laboratory interpretations

Introduction

Respiratory system is one of the systems of the human body where the exposure of the etiological factor in most of the occasions is through either nose or mouth; less frequently through heart via pulmonary capillaries and even less frequently through alimentary canal through reverse movement happening via oesophagus.

Depending upon the nature of aetiological factor, virulence of aetiological factor, number of aetiological factors entering into the respiratory system, nature of immune status of the human body, the clinical features may vary.

If the aetiological factors enter via nose, the clinical symptoms start from nose, nasopharynx and if the aetiological factors enter via oral cavity, initial symptoms start from oral cavity, larynx etc. if the aetiological factors enter into the respiratory passages through pulmonary capillaries, the initial symptoms start as cough with pink and frothy sputum.

Peculiarities of respiratory system and diagnosis of respiratory disease:

- ⊗ Respiratory system is a peculiar system where entry and exit of the aetiological factor is the same i.e. through nose. In other words, oxygen rich air and carbon dioxide rich air move via nose and nasal cavity.
- ⊗ Respiratory system can be divided into upper and lower respiratory system. Upper respiratory system is comprising of organs from nose to larynx and

lower respiratory system is comprising of organs from trachea to alveoli. Pleural cavity is considered as associated part of lower respiratory tract.

- ⊗ When upper respiratory tract is involved, an individual may have sneezing, blockage of nose and nasal cavity, imperception of smell perception, running nose, pain in the ears (due to blockage of eustachian tube), cough with or without expectoration, breathlessness during inspiration, post nasal drip, mouth breathing (due to nasal blockage) and hoarseness of voice. When the lower respiratory tract is afflicted, the person may have breathlessness during expiration, cough, expectoration of sputum, chest discomfort, haemoptysis. When an individual is having affliction of pleural space, he may have short cough with pain on sides of the chest while coughing.
- ⊗ When an individual is presented with known respiratory disease, sooner or later he may have affliction of right side of the heart (right ventricle or right atrium) which is being termed as cor pulmonale.

Pneumonia

Pneumonia (see Figure 31) is acute inflammation of the lungs caused by infection. Initial diagnosis is usually based on chest x ray. Causes, symptoms, treatment, preventive measures and prognosis differ depending on whether the infection is bacterial, viral, fungal or parasitic; whether it is acquired in the community, hospital or nursing home; whether it develops in a patient who is immunocompetent or immunocompromised.

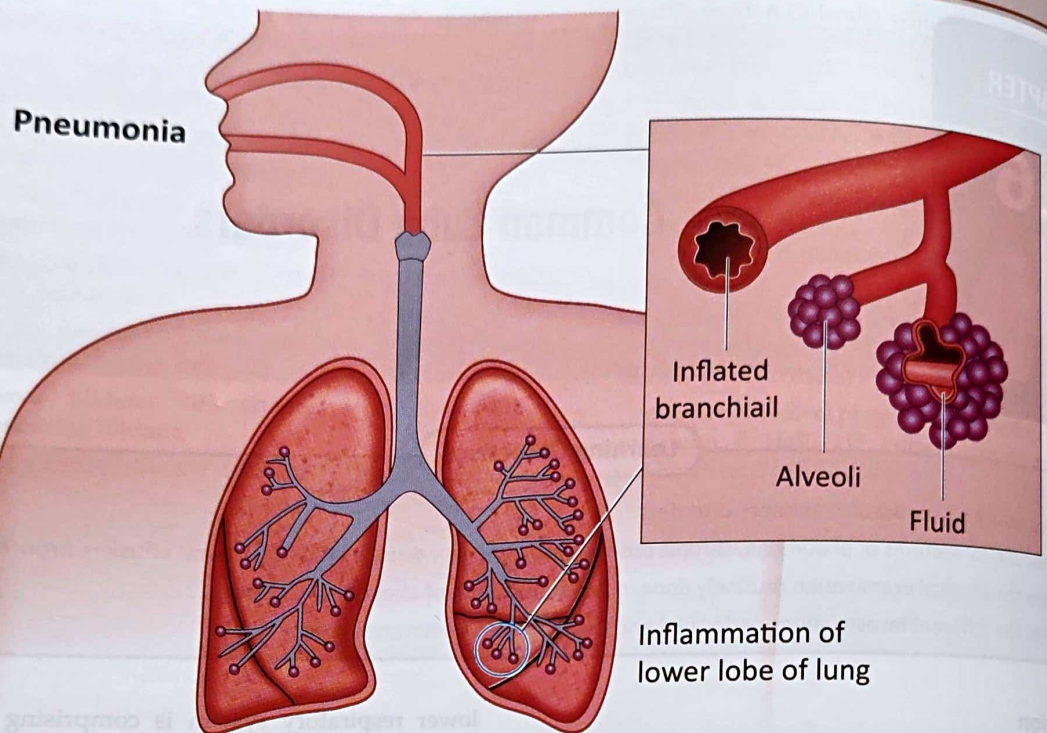


Figure 31: Mechanism of pneumonia

- Clinical features may vary depending upon the cause, setup where the infection has occurred in the patient and nature of the immunity of the patient who has acquired pneumonia.
- In community acquired pneumonia, symptoms and signs include productive cough, pleuritic chest pain, dyspnoea, tachypnoea and tachycardia. This may be associated with fever, bronchial breath sounds, crackles and dullness on percussion. Nasal flaring and use of accessory respiratory muscles may also be seen.
- In hospital acquired pneumonia, the clinical features are similar to those of community acquired pneumonia, but in ventilated patients, pneumonia may also manifest as worsening oxygenation and increased tracheal secretions.
- In nursing home acquired pneumonia, the clinical features are similar to the symptoms of other settings, but the mortality in nursing home acquired pneumonia are moderately high, but may be due to the presence of comorbidities.
- Diagnosis is done by chest x ray (see Figure 32) bronchoscopy, blood cultures, assessment of renal function and oxygenation, pulse oximetry. Chest x ray is the earliest diagnostic tool in the diagnosis of pneumonia and other investigation tools help in the cause of the disease and prognosis of the disease in the patients.

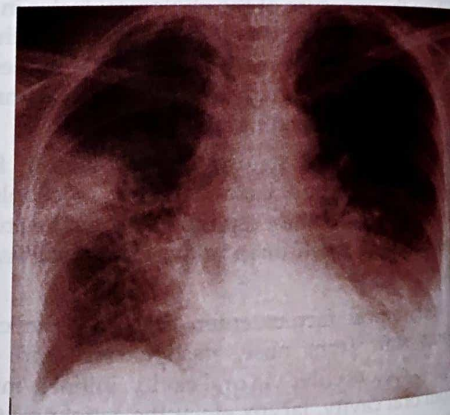


Figure 32: Chest X ray of patient of pneumonia

Chronic Obstructive Pulmonary Diseases (COPD)

Chronic Obstructive Pulmonary Disease (COPD) (Figure 33) is partially reversible airflow limitation caused by inflammatory response to inhaled toxins, of cigarette smoke. Symptoms generated are productive cough and breathlessness that develop over years. Signs include decreased breath sounds, prolonged expiratory phase of respiration and wheezing. Severe cases may have weight loss, pneumothorax, right heart failure, acute or chronic respiratory failure. Diagnosis is based on history, physical examination, chest x ray and pulmonary function tests. COPD comprises chronic obstructive bronchitis and emphysema.

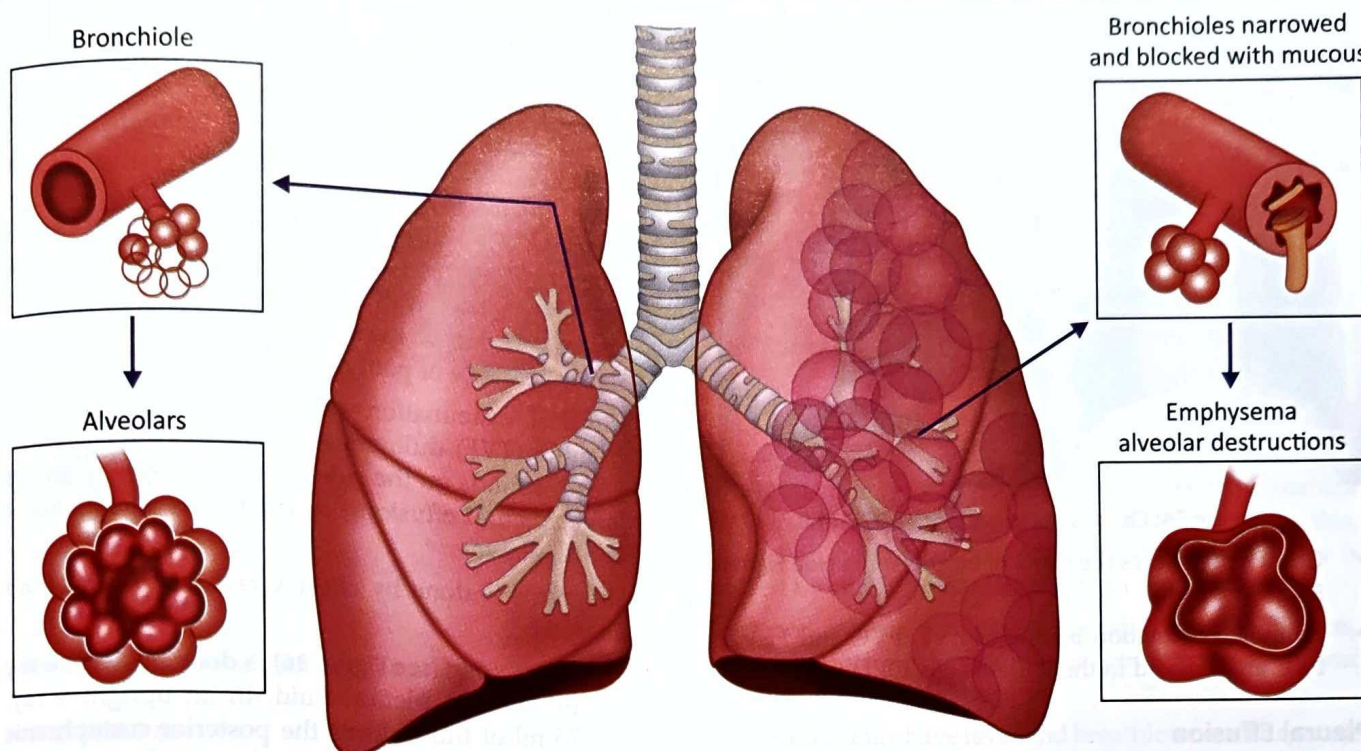


Figure 33: Mechanism of COPD

- Chronic obstructive bronchitis is chronic bronchitis with airflow obstruction. Chronic bronchitis is defined as productive cough on most days of the week for at least 3 months total duration for consecutive 2 years. Chronic bronchitis becomes chronic obstructive bronchitis if spirometric evidence of airflow obstruction develops. Chronic asthmatic bronchitis is a similar, overlapping condition characterized by chronic productive cough, wheezing and partially reversible airflow obstruction. It occurs predominantly in smokers, with a history of asthma.
- Emphysema is destruction of lung parenchyma leading to loss of elastic recoil and loss of alveolar septa and radial airway traction, which increases the tendency of airway collapse. Lung hyperinflation, airflow limitation and air trapping follow.
- Etiology of chronic obstructive pulmonary disease is smoking and genetic factors.

Pathology of COPD

- **Inflammation:** inhalation of cigarette smoke etc irritants triggers inflammatory response.
- **Infection:** this may occur along with inflammation, the most common cause is being bacteria, *Haemophilus influenzae* in particular.
- **Airflow limitation:** this is caused by airway obstruction, loss of elastic recoil or both.
- Complications of this pathological event are pulmonary hypertension, respiratory infection, weight loss.

Clinical features

- COPD takes years to develop and progress. Most patients have smoked ≥ 20 cigarettes/day for ≥ 20 years.
- Productive cough is the initial symptom, developing in smokers in their 40s or 50s.
- Breathlessness (dyspnoea) that is progressive, persistent, exertional or worse during respiratory infection, appears when patients are in their 40s or 50s.
- Symptoms usually progress quickly in patients who continue to smoke and in those who have a higher lifetime tobacco exposure.
- Morning headache develops in more advanced disease and signals nocturnal hypercapnia or hypoxemia.
- Signs of COPD include wheezing, increased expiratory phase of breathing, lung hyperinflation manifested as decreased heart and lung sounds and increased antero-posterior diameter of thorax (barrel shaped chest). In advanced cases of COPD, pursed-lip breathing, accessory respiratory muscles usage, paradoxical inward movement of lower intercostal interspaces during inspiration (Hoover's sign), central cyanosis. Signs of cor pulmonale may develop which include neck vein distension, S2 split with accentuated pulmonary component, tricuspid regurgitation and peripheral oedema.
- Spontaneous pneumothorax may develop as a result of rupture of bullae in the lungs.

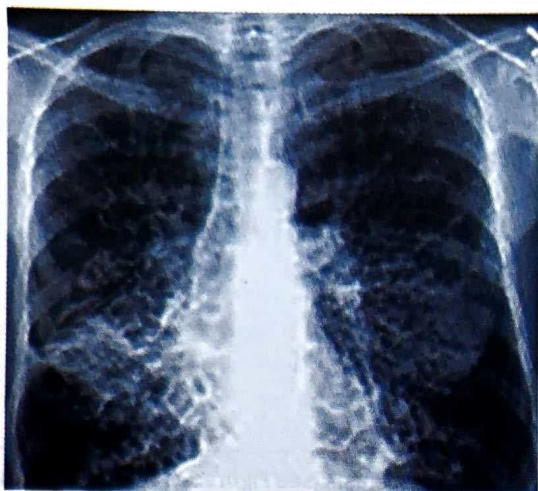


Figure 38: Chest X ray of patient of bronchiectasis

High resolution CT scan

100% sensitive and specific test for the diagnosis of bronchiectasis.

Pulmonary function tests

- ⊗ These may be helpful in documenting the baseline function and for following the progression of the disease over time.
- ⊗ Reduced FEV1 (Forced expiratory volume in 1 second), reduced FVC (Forced Vital Capacity) and FEV1/FVC are noted in this disease.

Key summary of the chapter

- ⊗ Respiratory system is a system of the human body where in oxygenated and deoxygenated air will move via the same passage across the human body and environment. This is further subdivided into two upper and lower respiratory system for localizing the lesion, prediction of prognosis and proper management of the disease and prevention of the disease too.
- ⊗ In this chapter, pneumonia, chronic obstructive pulmonary disease, pleural effusion and bronchiectasis are explained in terms of meaning, aetiology, pathology, clinical features in brief, method of examination in these patients and diagnosis of the diseases in brief along with necessary chest x ray findings.

Self-Assessment: Questions for Reviews

Short essay questions

- ⊗ Explain pneumonia
- ⊗ Explain chronic obstructive pulmonary disease
- ⊗ Explain pleural effusion
- ⊗ Explain bronchiectasis

MCQs

1. Which of the disease is responsible for the manifestation of cor pulmonale?
 - a. Coronary artery disease;
 - b. Chronic bronchitis;
 - c. Left ventricular failure;
 - d. Left atrial failure

Answer: (b)

2. Honeycombing appearance in the chest x ray is indicative of
 - a. Pneumonia;
 - b. Chronic obstructive pulmonary disease;
 - c. Pleural effusion;
 - d. Bronchiectasis

Answer: (d)

3. Meniscus sign in chest x ray is indicative of
 - a. Pneumonia;
 - b. Chronic obstructive pulmonary disease;
 - c. Pleural effusion;

- d. Bronchiectasis

Answer: (c)

4. The presenting complaint 'hoarseness of voice' is indicative of the affliction of
 - a. Larynx;
 - b. Trachea;
 - c. Nasopharynx;
 - d. Primary bronchi

Answer: (a)

5. 30 years old female patient is presenting with cough since 4 days, running nose since 3 days, nasal blockage since 2 days and pain in the ear on left side since 1 day. Pain in the left ear is indicative of
 - a. Rupture of ear ossicles on left side;
 - b. Rupture of tympanic membrane on left side;
 - c. Accumulation of wax in the left ear;
 - d. Blockage of left eustachian tube

Answer: (d)

6. Barrel shaped chest is present in the patients of
 - a. Pneumonia;
 - b. COPD;
 - c. Pleural effusion;
 - d. Bronchiectasis

Answer: (b)

7. The amount of fluid should be present in pleural cavity, if costophrenic angle on the chest x ray PA view should be obliterated is
- 10 ml;
 - 75 ml;
 - 175 ml;
 - 300 ml

Answer: (c)

8. Which of the following radiological sign is not indicative of the diagnosis of emphysema?
- Reduced cardiac shadow in chest PA view;
 - Flattened diaphragm;
 - Absent hilar shadows;
 - Meniscus sign

Answer: (d)

9. Which of the following radiological sign is indicative of the diagnosis of bronchiectasis?
- Flattened diaphragm;
 - Honeycombing;
 - Meniscus sign;
 - Bats wing appearance

Answer: (b)

10. Which of the following clinical condition where the individual may have breathlessness during inspiration?
- Bronchiectasis;
 - Bronchial asthma;
 - Impaction of foreign body in trachea;
 - Nasal polyps

Answer: (d)

Chapter resources

- <https://www.svhlunghealth.com.au/conditions/pneumonia>
- <https://www.fitpaa.com/blog/what-are-the-symptoms-of-chronic-obstructive-pulmonary-disease-copd/>
- https://en.wikipedia.org/wiki/Pleural_effusion
- <https://www.medicoverhospitals.in/diseases/bronchiectasis/>
- The Merck Manual of Diagnosis and Therapy by Robert S Porter and Justin L Kaplan, 19th edition, Merck Sharp & Dohme Corp, New Jersey, 2011
- Golwalla's medicine for students by Aspi F Golwalla and Sharukh A Golwalla, edited by Milind Y Nadkar, 25th edition, Jaypee Brothers Medical Publishers (P) Ltd, New Delhi, 2017
- Chamberlain's symptoms and signs in clinical medicine by Colin Ogilvie and Christopher C Evans, 12th edition, Butterworth-Heinemann Publishers, 1997
- Best & Taylor's physiological basis of medical practice by O P Tandon and Y Tripathi, 13th edition, Wolters Kluwer Health (India) Pvt. Ltd, Gurgaon, 2012
- Textbook of Medical physiology by Arthur C Guyton and John E Hall, 11th edition, Saunders Elsevier publications, Philadelphia, 2006
- www.chestmed.theclinics.com
- www.stritch.luc.edu
- www.rsnajournals.org
